

# PIZZA SALES SQL ANALYSIS

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**Tech Stack:**  
**MySQL | SQL | MySQL Workbench**

# Pizza Sales SQL Analysis

## Project Overview:

This project analyzes pizza sales data that focuses on analyzing pizza sales data using SQL. The project aims to extract meaningful business insights related to sales performance, customer ordering patterns, and product popularity.

The analysis was performed using MySQL by applying SQL queries involving data aggregation, joins, and window functions. The project covers revenue analysis, best-selling pizzas, order trends, and category-wise performance to understand business growth and customer preferences.

## Objectives:

The main objectives of this project are:

- To analyze pizza sales data using SQL and extract meaningful business insights.
- To calculate total orders and revenue generated from pizza sales.
- To identify the most popular pizza types and customer ordering preferences.
- To analyze sales performance based on pizza categories, sizes, and products.
- To determine the highest revenue-generating pizzas.
- To apply SQL concepts such as joins, aggregations, CTEs, and window functions for data analysis.
- To understand how data can be used for business decision-making.

## Dataset Description:

The dataset used in this project contains pizza sales transaction data. It includes information about customer orders, pizza details, quantities, prices, and categories.

The dataset was obtained from GitHub and consists of four relational tables that were analyzed using MySQL.

## Tables in Dataset

| Table Name    | Description                                                                        |
|---------------|------------------------------------------------------------------------------------|
| orders        | Contains order information such as order ID, order date, and order time.           |
| order_details | Contains details of pizzas included in each order along with the quantity ordered. |
| pizzas        | Contains pizza information such as pizza size, price, and pizza type ID.           |
| pizza_types   | Contains pizza names, categories, and ingredients.                                 |

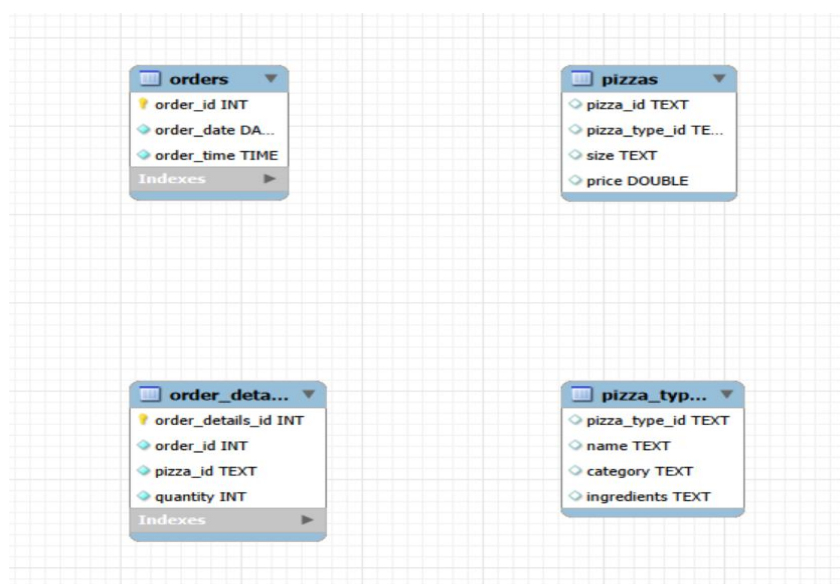
## Dataset Purpose:

The dataset helps in analyzing:

- Total sales and revenue
- Customer ordering patterns
- Popular pizza types
- Pizza category performance
- Revenue contribution of different products

## Database Schema:

The database consists of four relational tables. These tables are connected using primary keys and foreign keys to perform sales analysis.



## Schema Details

| Table         | Primary Key      | Important Columns            |
|---------------|------------------|------------------------------|
| orders        | order_id         | order_date, order_time       |
| order_details | order_details_id | order_id, pizza_id, quantity |
| pizzas        | pizza_id         | pizza_type_id, size, price   |
| pizza_types   | pizza_type_id    | name, category, ingredients  |

## Relationships

- One order can contain multiple pizza items.
- order\_details connects orders with pizzas using order\_id.
- pizzas connects with pizza types using pizza\_type\_id.
- Pizza information such as name and category is obtained from pizza\_types.

## Tools & Technologies:

| Category                 | Technology      |
|--------------------------|-----------------|
| Database                 | MySQL           |
| Query Language           | SQL             |
| Database Management Tool | MySQL Workbench |

## Description:

- **MySQL:** Used for storing, managing, and querying the pizza sales dataset.
- **SQL:** Used to perform data analysis using queries, joins, aggregate functions, CTEs, and window functions.
- **MySQL Workbench:** Used to create database tables, execute SQL queries, and analyze results.

## SQL Analysis Performed:

The project involved performing SQL-based analysis on pizza sales data to extract meaningful business insights. Various SQL queries were written using joins, aggregations, and window functions.

### **Basic Analysis**

- Retrieved the total number of orders placed.
- Calculated total revenue generated from pizza sales.
- Identified the highest-priced pizza.
- Found the most commonly ordered pizza size.
- Listed the top 5 most ordered pizza types based on quantity.

### **Intermediate Analysis**

- Analyzed total quantity of pizzas ordered by category.
- Determined order distribution based on hour of the day.
- Calculated category-wise distribution of pizzas.
- Found the average number of pizzas ordered per day.
- Identified the top 3 pizzas based on revenue.

### **Advanced Analysis**

- Calculated the percentage contribution of each pizza type to total revenue.
- Analyzed cumulative revenue growth over time.
- Identified the top 3 revenue-generating pizzas within each category using ranking functions.

### **SQL Techniques Used**

- SELECT statements
- Filtering and sorting
- Aggregate functions (SUM, COUNT, AVG)
- GROUP BY
- INNER JOIN
- Common Table Expressions (CTEs)

- Window Functions (RANK(), SUM() OVER())

## Business Questions Solved:

The project answers key business questions related to pizza sales performance and customer preferences:

### **Sales Performance**

- How many total orders were placed?
- What is the total revenue generated from pizza sales?
- Which pizza generates the highest revenue?
- What is the overall sales trend over time?

### **Customer Preferences**

- Which pizza size is ordered the most?
- Which pizza types are most popular among customers?
- What are the top-selling pizzas based on order quantity?

### **Product Analysis**

- Which pizza category receives the highest number of orders?
- Which pizza types contribute the most to total revenue?
- What are the top 3 revenue-generating pizzas within each category?

### **Revenue Analysis**

- What percentage does each pizza type contribute to total revenue?
- How does cumulative revenue increase over time?

## Key Insights & Findings:

The analysis of pizza sales data provided the following insights:

### **Sales Overview**

- A total of **21,350 orders** were placed during the analyzed period.
- The total revenue generated from pizza sales was **\$817,860.05**.

### **Customer Ordering Pattern**

- The most preferred pizza size was **Large (L)** with **18,526 orders**.

- Large-sized pizzas had significantly higher demand compared to XL and XXL sizes.

### Popular Pizza Analysis

- The most ordered pizza was **The Classic Deluxe Pizza** with **2,453 orders**.
- Other highly ordered pizzas included:
  - The Barbecue Chicken Pizza (2,432 orders)
  - The Hawaiian Pizza (2,422 orders)
  - The Pepperoni Pizza (2,418 orders)

### Revenue Analysis

- The highest-priced pizza was **The Greek Pizza** with a price of **\$35.95**.
- The top revenue-generating pizzas were:
  - **The Thai Chicken Pizza** — \$43,434.25
  - **The Barbecue Chicken Pizza** — \$42,768.00
  - **The California Chicken Pizza** — \$41,409.50

### Business Insights

- Customer demand is mainly concentrated around a few popular pizza varieties.
- Large pizzas contribute significantly to sales volume.
- Revenue analysis helps identify products that are most valuable for business growth.

## Conclusion:

This project demonstrates the use of SQL for analyzing pizza sales data and extracting meaningful business insights. By using SQL queries, joins, aggregate functions, and window functions, the project analyzed sales performance, customer preferences, and revenue trends.

The analysis helped identify important metrics such as total orders, total revenue, best-selling pizzas, popular sizes, and top revenue-generating products.

This project improved practical understanding of database querying and data analysis techniques, showing how SQL can be used to support data-driven business decisions.

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